

Homework 6: When Does Utility Maximization Have a Solution?

GSE 510

Instructions: Answer all parts carefully. The goal of this assignment is to think about when a consumer's utility maximization problem actually has a solution.

Consider a consumer who chooses a bundle

$$x = (x_1, x_2) \in \mathbb{R}_+^2$$

from the budget set

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq m\},$$

where $p_1 > 0$, $p_2 > 0$, and $m > 0$. The consumer's problem is

$$\max_{x \in B} u(x).$$

Question 1. The Budget Set

(10 points)

- (a) Show that B is nonempty.
- (b) Explain why B is closed.
- (c) Explain why B is bounded.
- (d) Draw the budget set in $\mathbb{R}_+^2$.
- (e) If the budget set is closed and bounded, what does this suggest about the “size” of the set of feasible choices?

Question 2. A Well-Behaved Utility Function

(10 points)

Suppose

$$p_1 = p_2 = 1, \quad m = 1,$$

so that the budget set is

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 1\}.$$

First consider the utility function

$$u(x_1, x_2) = x_1 + x_2.$$

- (a) What is the largest possible value of $x_1 + x_2$ on the budget set?
- (b) Find all bundles in B that maximize $u(x_1, x_2) = x_1 + x_2$.
- (c) Does a utility-maximizing bundle exist?
- (d) Is the utility function $u(x_1, x_2) = x_1 + x_2$ continuous?

Question 3. A Utility Function with a Jump

(10 points)

Now use the same budget set

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 1\}.$$

Consider the utility function

$$u(x_1, x_2) = \begin{cases} x_1, & \text{if } x_1 < 1, \\ 0, & \text{if } x_1 = 1. \end{cases}$$

- (a) Can the consumer choose bundles with x_1 arbitrarily close to 1?
- (b) What happens to $u(x_1, x_2)$ as x_1 gets closer and closer to 1 from below?
- (c) What is the value of utility at the bundle $(1, 0)$?
- (d) Does this utility function attain a maximum on B ?
- (e) What goes wrong in this example?

Question 4. Another Failure: A Missing Limit Value

(10 points)

Again let

$$B = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1 + x_2 \leq 1\}.$$

Consider the utility function

$$u(x_1, x_2) = \begin{cases} x_1, & \text{if } x_1 < 1, \\ \frac{1}{2}, & \text{if } x_1 = 1. \end{cases}$$

- (a) What is the largest value that utility gets close to as $x_1 \rightarrow 1$ from below?
- (b) Is that value actually achieved by any feasible bundle?
- (c) Does the consumer have a utility-maximizing bundle?
- (d) Is the problem caused by the budget set or by the utility function?
- (e) Explain in words why a jump or discontinuity can prevent a maximizer from existing.

Question 5. What Condition Do We Need?

(10 points)

Based on the examples above, answer the following.

- (a) Is it enough for the budget set to be closed and bounded? Explain.
- (b) What additional property should the utility function have so that utility values do not “jump” at limit points?
- (c) Give one example of a continuous utility function on B that has more than one maximizer.